1.

quadratic.h

```h
class Quadratic
{
    private:
        double a,b,c;

    public:
        Quadratic( double x, double y , double z );
        void print();
};
```

quadratic.cpp

```cpp
#include"quadratic.h"
#include<iostream>
#include<cmath>
using namespace std;

Quadratic::Quadratic( double x , double y , double z ) : a(x) , b(y) , c(z)
{
}

void Quadratic::print()
{
    cout << a << "x^2+" << b << "x+" << c << endl;

    //
    double delta = b*b - 4*a*c;


    if( delta>0 )
    {
        double root1 = (-b + sqrt(delta)) / (2*a);
        double root2 = (-b - sqrt(delta)) / (2*a);
        cout << "Root1: " << root1 << " Root2: " << root2 << endl;
    }

    else if ( delta==0 )
    {
        double root = -b / (2*a);
        cout << "Root1: " << root << "  Root2: " << root << endl;
    }
    else
    {
        cout << "The equation has no roots." << endl;
    }

}
```

quadraticMain.cpp

```cpp
#include"quadratic.h"
#include<iostream>
using namespace std;

int main ()
{
    Quadratic quad1 (2,3,1);
    quad1.print ();
    cout << endl;

    Quadratic quad3 (1,4,4);
    quad3.print();
    cout << endl;

    Quadratic quad4 (2,2,4);
    quad4.print();

    return 0;
}
```

```
2x^2+3x+1
Root1: -0.5 Root2: -1

1x^2+4x+4
Root1: -2  Root2: -2

2x^2+2x+4
The equation has no roots.

--------------------------------
Process exited after 0.2303 seconds with return value 0
請按任意鍵繼續 . . .
```

2.

zeller.h

```
class Zeller
{
    private:
        int month,day,year;

    public:
        Zeller( int m , int d , int y );
        void print();
};
```

zeller.cpp

```
#include"zeller.h"
#include<iostream>
using namespace std;

Zeller::Zeller( int x , int y , int z ) : month(x) , day(y) , year(z)
{
}

void Zeller::print()
{
    int weekday = ( day + 26*(month+1)/10 + year + year/4 - year/100 + year/400 ) %7;

    cout << month << "/" << day << "/" << year << " is ";
    switch( weekday )
    {
        case 1:
            cout << "Sunday";
            break;
        case 2:
            cout << "Monday";
            break;
        case 3:
            cout << "Tuesday";
            break;
        case 4:
            cout << "Wednesday";
            break;
        case 5:
            cout << "Thursday";
            break;
        case 6:
            cout << "Friday";
            break;
        case 7:
            cout << "Saturday";
            break;

    }

}
```

[*] zellerMain.cpp

```cpp
#include"zeller.h"
#include<iostream>
using namespace std;

int main ()
{

    // Instantiation and printing weekday for first object
    Zeller z1 (3,29,2016);
    z1.print();
    cout << endl;

    // Instantiation and printing weekday for second object
    Zeller z2 (5,24,2015);
    z2.print();
    cout << endl;

    // Instantiation and printing weekday for third object
    Zeller z3 (8,7,2017);
    z3.print();

    return 0;
}
```

```
3/29/2016 is Tuesday
5/24/2015 is Sunday
8/7/2017 is Monday
--------------------------------
Process exited after 0.2144 seconds with return value 0
請按任意鍵繼續 . . .
```

3.

complex.h

```
class Complex
{
    private:
        double real,imag;

    public:
        Complex( double x , double y );

        void addTo( Complex other );
        void subtractFrom( Complex other );
        void multiplyBy( Complex other );
        void divideBy( Complex other );

        void print();
};
```

complex.cpp

```
#include"complex.h"
#include<iostream>
#include<iomanip>
using namespace std;

Complex::Complex( double x , double y ) : real(x) , imag(y)
{
}


void Complex::addTo( Complex other )
{
    real = real + other.real;
    imag = imag + other.imag;
}
void Complex::subtractFrom( Complex other )
{
    real = real - other.real;
    imag = imag - other.imag;
}
```

```
// (a+bi)*(c+di) = (ac-bd)+(ad+bc)i
void Complex ::multiplyBy( Complex other )
{
    double newReal = real * other.real - imag * other.imag;
    double newImag = real * other.imag + imag * other.real;
    real = newReal;
    imag = newImag;
}
// (a+bi)/(c+di) = ( (ac+bd)+(bc-ad)i ) / ( c*c+d*d )
void Complex::divideBy( Complex other )
{
    double denominator = other.real * other.real + other.imag * other.imag;
    if( denominator==0 )
    {
        cout << "Error: Division by zero." << endl;
        return;
    }
    else
    {
        double newReal = ( real * other.real + imag * other.imag ) / denominator;
        double newImag = ( imag * other.real - real * other.imag ) / denominator;
        real = newReal;
        imag = newImag;
    }
}
```

```


void Complex::print()
{
    cout << "(" << showpos << fixed << setprecision(2) << real << ")+i(" << imag << noshowpos << ")" << endl;
}

```

complexMain.cpp

```cpp
#include "complex.h"
#include<iostream>
using namespace std;

int main ()
{
    // Instantiation of four complex numbers
    Complex complex1 (2.0, 3.0);
    Complex complex2 (3.0, 4.0);
    Complex complex3 (5.0, 4.0);
    Complex complex4 (6.0, 5.0);

    // Instantiation a complex number to be used in operation
    Complex temp (2.0, 1.0);

    // Applying operations on the four complex numbers
    complex1.addTo (temp);
    complex2.subtractFrom (temp);
    complex3.multiplyBy (temp);
    complex4.divideBy (temp);

    // Printing the result of operations
    cout << "complex1 + temp = ";
    complex1.print ();
    cout << "complex2 - temp = ";
    complex2.print ();
    cout << "complex3 * temp = ";
    complex3.print ();
    cout << "complex4 / temp = ";
    complex4.print ();
    return 0;
}
```

```
complex1 + temp = (+4.00)+i(+4.00)
complex2 - temp = (+1.00)+i(+3.00)
complex3 * temp = (+6.00)+i(+13.00)
complex4 / temp = (+3.40)+i(+0.80)

--------------------------------
Process exited after 0.04582 seconds with return value 0
請按任意鍵繼續 . . .
```